Mid Semester Examination 2025

Name of the Program: M. Tech.(Electric Vehicle Technology) *Semester: First*

Name of the Course: Research Methodology and IPR *Course Code:* **MEV 106**

Time: 1-1/2 Hour Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing ***any one of the sub questions***.

(ii) Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	Explain the meaning of a research problem. Discuss the different sources from which a research problem can be identified, and illustrate with suitable examples.	
	OR	
(b)	What are the essential criteria and characteristics of a good research problem? Why is the proper formulation of a research problem important in ensuring quality research outcomes?	
Q2	(10 marks)	CO2
(a)	Identify and explain common errors in selecting a research problem. How can a researcher avoid these mistakes while defining the scope and objectives of the research?	
	OR	
(b)	Discuss the various approaches available for investigating solutions to a research problem. Highlight how data collection, analysis, and interpretation play a key role in these approaches.	
Q3	(10 marks)	CO1
(a)	Describe the importance of necessary instrumentation in conducting research. Explain with examples how proper selection of tools and techniques influences the accuracy, reliability, and interpretation of research findings.	
	OR	
(b)	What are the effective approaches for conducting a literature study? Explain how proper literature review and analysis help in identifying research gaps and framing a strong research problem.	
Q4	(10 marks)	CO2
(a)	Define plagiarism. Discuss different types of plagiarism, their consequences, and the strategies researchers can adopt to avoid plagiarism in academic writing.	
	OR	
(b)	What is research ethics? Explain the importance of ethical practices in conducting research, with examples from data collection, authorship, and publication.	CO3
Q5	(10 marks)	CO2
(a)	Explain the key principles of effective technical writing. How does technical writing differ from general writing? Discuss the steps involved in writing a good research report.	
	OR	
(b)	What are the essential components of a research proposal? Explain the format of a standard research proposal and the guidelines for developing it effectively.	CO3